Astaxanthin and Crocin Acid Experimental Record

Notebook Astaxanthin and Crocetin

July to September 2026

We conducted plasmid construction and transformation experiments focusing on the synthesis of astaxanthin and crocin acid in *Dunaliella salina*, targeting related genes. The astaxanthin module primarily involved *HpBKT*, while the crocin acid module mainly involved *GjCCD4a* and *GjALDH2C3*. Using pCAMBIA1302 as the vector, we sequentially performed fragment amplification, homologous recombination, bacterial clone screening, and sequencing, and attempted to introduce the constructed plasmids into *Dunaliella salina*.

As the experiment progressed, we found that some candidate plasmids might have incomplete expression cassettes, so we proceeded to reconstruct and re-screen them. Below, we document our work, observations, and adjustment process in chronological order, along with corresponding electrophoresis gels, DNA concentration data, and sequencing results.

July 27 to August 2: Experimental preparation and vector amplification

This week, we mainly prepared the media, amplified the empty vector, and set up the experimental materials needed for subsequent cloning and *Dunaliella* cultivation.

On July 27, we prepared LB liquid medium and LB solid medium containing kanamycin, completed sterilization, and poured the plates. After the plates solidified, we labeled them and stored them in the refrigerator, preparing for subsequent bacterial transformation and single colony screening.

From July 28 to July 30, after verifying the pipetting procedure, we transformed pCAMBIA1302 into DH5α competent cells. After colonies appeared on selective plates, we selected well-growing colonies for liquid culture amplification, followed by plasmid extraction. The concentration was not measured during this extraction; the plasmid was stored at low temperature for subsequent vector processing.

From July 31 to August 2, we prepared the DM medium stock solution and culture medium required for *Dunaliella salina* cultivation, and waited for the synthetic genes and primers to arrive.

August 3 to August 9: Fragment amplification and initial plasmid construction

This week, we began amplifying the vector backbone and the target gene, followed by homologous recombination, competent cell transformation, and screening of candidate clones.

From August 3 to August 5, we continued waiting for the gene and primers while attempting to amplify the pCAMBIA1302 backbone. The backbone PCR on August 4 was unsuccessful, so we repeated the PCR on August 5 and checked the amplified product by agarose gel electrophoresis.

On August 6, we amplified the fragments of *GjALDH2C3*, *GjCCD4a*, and *RBCS2*, performed gel electrophoresis, and recovered the DNA. The concentrations and absorbance ratios ofthe gel-recovered products for *GjALDH2C3* and *GjCCD4a* are shown in Table 1.

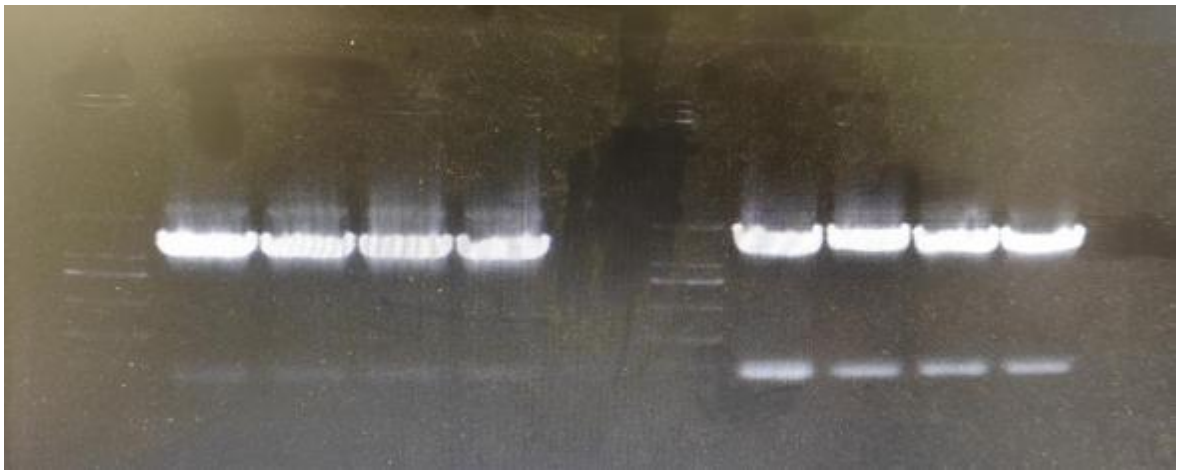


Figure 1. PCR electrophoresis results of *GjALDH2C3* and *GjCCD4a* fragments on August 6. The left lane shows the *GjALDH2C3* sample, the right lane shows the *GjCCD4a* sample, with DNA marker indicated beside each group.

Table 1. DNA fragment gel recovery data on August 6

Fragment	Concentration ng/μL	A260/A280	A260/A230
GjALDH2C3	152.20	1.882	1.268
GjCCD4a	138.20	1.900	2.069

On August 7, we performed homologous recombination between the linearized pCAMBIA1302 and the *GjALDH2C3* and *GjCCD4a* fragments, respectively, and transformed the assembled products into Trelief5α competent cells. On the same day, we also amplified the *HpBKT* and *2146*-related fragments and continued with the corresponding assembly and transformation. The assembly reactions were carried out at 50°C for 30 minutes.

On August 8 to 9, after single colonies appeared on the transformation plates, we picked the colonies for liquid culture amplification. On August 9, we performed PCR, electrophoresis, and plasmid extraction on candidate clones related to *HpBKT*, *2146*, *GjALDH2C3*, and *GjCCD4a*, preparing them for further sequencing verification.



Figure 2. PCR electrophoresis results ofthe first-round candidate clones on August 9. Samples include constructs related to *2146*, *GjCCD4a*, *GjALDH2C3*, and *HpBKT*; DNA marker is located in the center.

Table 2 Plasmid concentration extracted on August 9

Candidate construction	Sample 1 ng/μL	Sample 2 ng/μL	Sample 3 ng/μL
1302-*HpBKT*	233.70	215.25	271.05
1302-*2146*	405.95	314.10	352.15
1302-*GjALDH2C3*	226.20	346.80	170.00
1302-*GjCCD4a*	341.40	334.70	261.40

August 10 to August 16: Sequencing preparation and adjustment of transformation conditions

This week, we sent the first round of candidate plasmids for sequencing and continued amplifying the vector and related constructs. The colonies on the transformation plates were too dense, prompting us to adjust the transformation and plating conditions.

From August 10 to August 13, we will send the plasmids extracted in the previous round for sequencing and await the results, while continuing with competent cell transformation, liquid culture amplification, and plasmid preparation for pCAMBIA1302 and pCAMBIA1302-*2146*.

From August 14 to August 15, we re-transformed the plasmids related to *HpBKT*, *GjALDH2C3*, and *GjCCD4a*. Since colonies on some plates were too dense for picking individual clones, we reduced the inoculation volume and repeated the transformation, while also setting up negative controls without plasmid to evaluate the effects of transformation conditions and antibiotic selection. We also attempted PCR amplification of the *Bar*-related fragment, but the electrophoresis bands did not match expectations.

On August 16, after adjustment, we were able to pick single colonies and proceed with liquid amplification. We used a smaller amount of competent cells for subsequent transformation, while continuing PCR for the *Bar*-related backbone, gel purification, and double digestion of pCAMBIA1302 with MfeI-HF and BstEII-HF.

August 17 to August 23: Initiation of constructs related to promoters and signal peptides

This week, we continued preparing plasmids and carried out new assembly and screening efforts focusing on *RBCS2* and signal peptide-related fragments. We also attempted to use the plasmid containing the target gene for transformation in *Dunaliella salina*.

On August 17, we extracted the plasmids related to *HpBKT*, *GjALDH2C3*, and *GjCCD4a*, measured their DNA concentration and absorbance ratios, and prepared materials for subsequent experiments.

Table 3 Plasmid extraction data on August 17

Candidate construction	Concentration ng/μL	A260/A280	A260/A230
1302-*HpBKT*	72.200	1.849	3.209
1302-*GjALDH2C3*	75.800	1.886	3.339
1302-*GjCCD4a*	74.850	1.739	4.101

On August 18, we picked single colonies from the transformation plates related to *RBCS2* and performed liquid amplification. Usable colonies were found on plates associated with *HpBKT*, *GjCCD4a*, and *GjALDH2C3*, while fewer colonies were observed for *2146*. We continued cultivation and arranged for the corresponding samples to be sent for sequencing.

On August 19, we extracted the *RBCS2*-related candidate plasmids and amplified the *GjALDH2C3*, *HpBKT*, *GjCCD4a* genes along with their promoter-related linker fragments using SP-specific primers, followed by gel electrophoresis and gel purification. Some of the purified products showed low A260/A230 ratios, prompting us to continue monitoring sample quality. On the same day, we also attempted to use the target gene-containing plasmids for transformation into *Dunaliella salina*.

Table 4 Concentration of recycled gel product on August 19

Segment Name	Concentration ng/μL
SP-*GjALDH2C3*	221.10
SP-*HpBKT*	213.35
SP-*GjCCD4a*	225.20
GjALDH2C3 sgp	227.85
HpBKT sgp	225.90
GjCCD4a sgp	170.45

SP and sgp are the fragment tags we used in this round of amplification.

From August 21 to August 22, after single colonies appeared on the plates, we picked the colonies for amplification and performed colony PCR on candidate constructs related to *RBCS2* and signal peptides, including *HpBKT*, *GjALDH2C3*, and *GjCCD4a*. We used the *RBCS2*-F primer together with the reverse primer specific to each target gene to verify the amplified fragments, followed by electrophoresis, plasmid extraction, and sequencing submission. The plasmid concentration obtained in this round was relatively low; data are shown in Table 5.

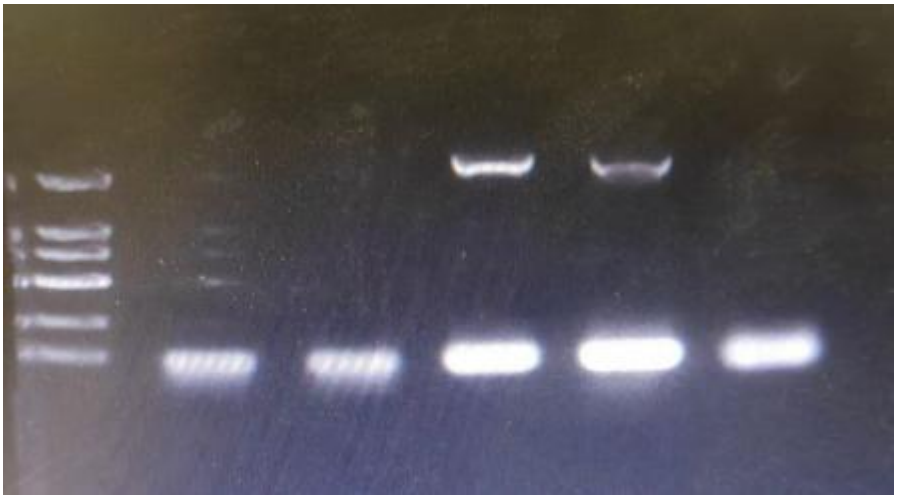


Figure 3. PCR electrophoresis results of bacterial cultures containing candidate constructs related to promoters and signal peptides, initiated on August 22. The left lane shows the DNA marker; the remaining lanes represent amplified products from the candidate clones.

Table 5 Concentration of candidate plasmids on August 22

Candidate construction	Sample 1 ng/μL	Sample 2 ng/μL
1302-*RBCS2*-sp-*HpBKT*	10.800	15.200
1302-*RBCS2*-sp-*GjALDH2C3*	13.800	15.050
1302-*RBCS2*-sp-*GjCCD4a*	14.735	16.040

sp stands for signal peptide, referring to the segment associated with the signal peptide.

August 24 to August 30: Reassembly and candidate plasmid preparation

This week, we continued repeating the purposeful fragment amplification and assembly, cultivating new candidate clones, and preparing plasmids required for the next round of validation.

From August 24 to August 25, we amplified the *HpBKT* fragment again, performed gel electrophoresis and gel extraction. Subsequently, we proceeded with homologous recombination and competent cell transformation for *RBCS2* with *HpBKT*, *GjCCD4a*, and *GjALDH2C3* fragments, while also amplifying the plasmid related to the 1302-*2146* region.

On August 26 to 27, after single colonies appeared on the transformation plates, we picked the colonies for liquid culture amplification and extracted plasmids related to 1302-*2146*. On August 27, we extracted candidate plasmids associated with *RBCS2-HpBKT*, *RBCS2-GjCCD4a*, and *RBCS2-GjALDH2C3*, and measured their concentrations as follows.

Table 6 Concentration of candidate plasmids on August 27

Candidate construction	Sample 1 ng/μL	Sample 2 ng/μL
1302-*RBCS2-HpBKT*	109.75	62.65
1302-*RBCS2-GjCCD4a*	94.550	109.35
1302-*RBCS2-GjALDH2C3*	119.05	125.95

From August 28 to August 30, we attempted to use 1302-*2146* for *Dunaliella* transformation while continuing to amplify the pCAMBIA1302 empty vector. Subsequently, we picked single colonies for liquid culture in preparation for the next round of plasmid extraction and sequencing.

August 31 to September 6: Initiation of suspected deletion and reconstruction

This week, we sent a new batch of candidate plasmids for sequencing. The sequencing results indicated that the previously picked recombinant plasmids might lack the promoter sequence of the target gene, prompting us to re-examine the construction and screening process of the expression cassette.

On August 31 to September 1, we extracted the previously cultured candidate plasmids and sent them for sequencing on September 1. During subsequent checks, we found that some of the candidate plasmids might not have retained the expected promoter sequence, so they need to be reconstructed, and the junctions between the promoter, signal peptide, and target gene must be verified.

From September 2 to September 3, we re-performed linearization ofpCAMBIA1302 and PCR amplification of the promoter-related fragments, followed by homologous recombination of the promoter, signal peptide, and fragments related to *HpBKT*, *GjCCD4a*, and *GjALDH2C3*, and completed competent cell transformation and plating.

On September 4, we performed PCR and gel electrophoresis on a new round of candidate colonies. The amplified bands were not of the expected size but instead resembled the target gene fragment without the promoter and signal peptide. This result suggests that the issue with the expression cassette assembly has not yet been resolved, and further reconstruction and screening are required.

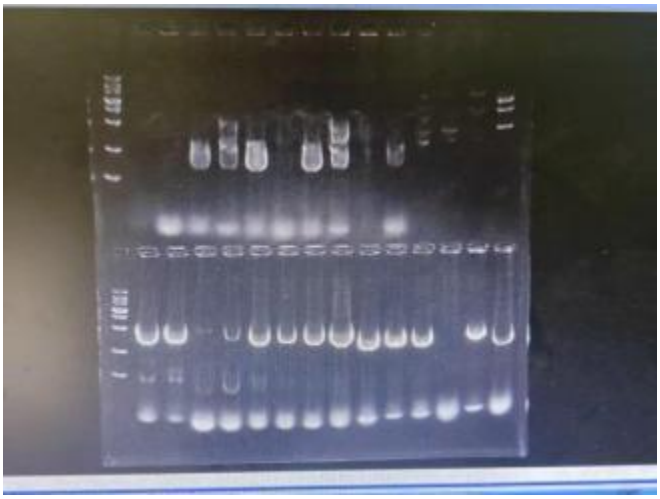


Figure 4. Colony PCR electrophoresis results on September 4. The first row, first 10 lanes, are samples related to *HpBKT*; the second row, first 8 lanes, are samples related to *GjCCD4a*, and the last 6 lanes are samples related to *GjALDH2C3*.

September 7 to September 27: Continued restructuring and material preparation

We will continue advancing plasmid reconstruction and candidate clone screening focusing on the suspected promoter deletion, with validation of the complete expression cassette as a key priority for subsequent work.

On September 22, we picked isolated single colonies from the p1302 empty vector plate and inoculated them into LB liquid medium containing kanamycin. The cultures were grown overnight at 37°C and 200 rpm, and then proceeded to prepare the vector material.

Resequencing and re-evaluation of candidate samples from September 28 to September 30

After reconstruction, we resubmitted the sample for testing on September 30 to verify whether the new candidate plasmid retained the target gene and to further confirm the integrity of the expression cassette.

The sequencing results we received in this round include P1302-*RBCS2*-4a, P1302-*RBCS2*-C3, and other candidate samples related to *GjCCD4a* and *GjALDH2C3*. The full plasmid consensus sequences, read alignments, and variant information from these results provide a basis for further examination of the candidate plasmids.

Sequence alignment revealed a complete reference coding region of *GjCCD4a* in P1302-*RBCS2*-4a and a complete reference coding region of *GjALDH2C3* in P1302-*RBCS2*-C3. These results suggest that both

candidate samples contain intact target genes; however, we still need to confirm that the promoters, signal peptides, insertion orientation, and junction regions all conform to the design.

Table 7 Consensus sequence length and target coding region alignment results returned by the end of September

Sample	Consensus sequence length bp	Purpose: Coding region alignment
P1302-*RBCS2*-4a	11,414	*GjCCD4a* full match
P1302-*RBCS2*-C3	11,258	*GjALDH2C3* full match
P1302-4a-1	10,549	No complete reference coding region match detected
P1302-4a-2	10,083	*GjCCD4a* full match
P1302-4a-3	10,083	*GjCCD4a* full match
P1302-4a-4	10,549	No complete reference coding region match detected
P1302-4a-5	10,139	*GjCCD4a* full match
P1302-4a-6	10,262	A single-base difference exists at the end of the coding region.
P1302-C3-1	8,875	Consensus indicates a single-base deletion in the coding region
P1302-C3-2	8,876	*GjALDH2C3* full match
P1302-C3-3	10,549	No complete reference coding region match detected
P1302-C3-4	10,549	No complete reference coding region match detected
P1302-C3-5	10,549	No complete reference coding region match detected
P1302-C3-6	10,549	No complete reference coding region match detected

Complete match refers to the full reference coding region of *GjCCD4a* or *GjALDH2C3* being identical at every base, which does not necessarily equate to successful validation of a complete expression cassette. The sequencing batch is SD260929184704, and the complete sample number can be found in the corresponding FASTA file.

We also compared sequence differences among the candidate samples. The consensus of P1302-C3-1 indicated a missing T at position 126 in the *GjALDH2C3* coding region, which requires re-examination of the original reads and trace files. In P1302-4a-6, a difference from A to G was observed at position 1994 in the *GjCCD4a* coding region, changing the stop codon from TAA to TGA—both of which are termination codons. We will use these differences to further select and validate candidate clones.

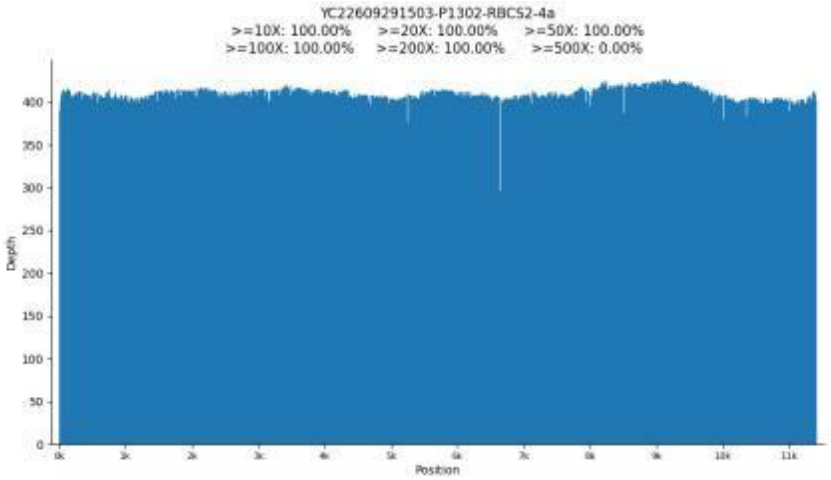


Figure 5. Raw sequencing coverage plot of P1302-*RBCS2*-4a. The x-axis represents the consensus sequence position, and the y-axis represents read depth.